

Wing Area to Wing Length Ratio

Semion D

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1 Introduction

Let's say we have 2 paper airplanes, one of them is really long, but at the same time narrow, and on the other hand we have a short but wide airplane. Both of them have the same wing area. The question is: what will fly longer and over longer distances, and what will be more effective? This will be a problem to solve in this short essay. It will be divided into 3 main parts: Hypothesis, which will make a prediction of the outcome based on math, physics and logic; next is an experiment, which will check the hypothesis; and the last one is the conclusion, which will summarize everything into a simple logic-math statement.

2 Hypothesis

2.1 Forces

At first we need to define the forces that act upon a complex airplane-looking object.

Here is the list:

Force	Representation
Gravity	Representation through constant $g = 9.81$ and mass of the object
Lift	Representation in Newtons as a variable L in formulas
Drag	Representation in Newtons as a variable D in formulas
Thrust	Representation in Newtons as a variable T in formulas

In this case we will not use thrust for several reasons: it does not affect the efficiency of the airplane itself, it just creates a continuous kinetic energy input; and it is harder to simulate on a small experimental scale. The only way where thrust will be included is the power input when we throw the airplane. So what's left is Gravity (or weight), Lift, and Drag.

Lift is a force that pushes the plane upwards. It happens because the area under the wing has higher air pressure than the top, and this pressure difference moves the airplane up. It is calculated through this formula:

$$L = \frac{1}{2}\rho v^2 SC_L$$

Variable	Meaning	Unit
L	Lift	Newtons (N)
ρ	Air density	kg/m ³
v	Air velocity	m/s
S	Wing area	m ²
C_L	Coefficient of lift	Scalar

C_L is a scalar from 0 to 1 and combines in itself many wing parameters, one of the main of them is an angle of attack of the wing α and it represents the angle of the wing. Here is the formula for it:

$$C_L = C_{L0} + a\alpha$$

a is the lift-curve slope, and in general it tells us the speed at which C_L grows when α grows. Basically it is the derivative of a linear function C_L . C_{L0} is the lift coefficient at 0° angle of attack, which might be positive or negative, but in this case we will consider this equal to 0.

To calculate lift-curve slope we can use this formula:

$$a = \frac{a_0}{1 + \frac{a_0}{\pi e AR}}$$

where:

a = lift-curve slope

a_0 = 2D airfoil lift-curve slope

e = Oswald efficiency factor

AR = aspect ratio

a_0 tells you how quickly the lift coefficient C_L changes with angle of attack for an idealized wing cross-section extending infinitely in the spanwise direction (was not expecting that i will write smth so clever). But counting that wing is a finite region, air starts to behave differently, creating friction and drag, which is included and considered in a . For our project a_0 will be:

$$a_0 \approx 2\pi \text{ rad}^{-1}$$

Next one is e which tells us how efficiently a real finite wing produces lift compared with an ideal wing. It is scalar from 0 to 1 and usually lies between 0.7 and 1, for this experiment we will chose 0.8, which is usual for flat wings.

Our plane will also experience drag, which we can not ignore, so here is the formula:

$$C_D = C_{D_p} + C_{D_i}$$

$$C_D = C_{D_p} + \frac{C_L^2}{\pi e AR}$$

And consequently:

$$D = \frac{1}{2} \rho v^2 S \left(C_{D_p} + \frac{C_L^2}{\pi e AR} \right)$$

Variable	Meaning	Unit
D	Total drag force	N
C_D	Total drag coefficient	Dimensionless
ρ	Air density	kg/m ³
v	Air velocity	m/s
S	Wing area	m ²
C_{D_p}	Parasite/profile drag coefficient	Dimensionless
C_L	Lift coefficient	Dimensionless
e	Oswald efficiency factor	Dimensionless
AR	Wing aspect ratio	Dimensionless

Before actually calculating parameters of our 2 prototypes we need to establish the difference between them using math. Engineers did it already long time ago and came up with this formula:

$$AR = \frac{b^2}{S}$$

AR is the aspect ratio, which represents the relationship between wing area S and wing span b which is a distance from one tip of the wing to another. This formula will help us to separate 2 prototypes on a mathematical level.

2.2 Calculation of Characteristics for Wide Airplane

We need to remember that we have 4 main forces acting upon this object.

$$\vec{L} = \mathbf{Lift}$$

$$\vec{D} = \mathbf{Drag}$$

$$\vec{T} = \mathbf{Thrust}$$

$$\vec{W} = \mathbf{Gravity}$$

We know that $\vec{L}$ is perpendicular to $\vec{D}$ and $\vec{T}$, but parallel to $\vec{W}$ so:

$$\vec{L} \perp \vec{D}$$

$$\vec{L} \perp \vec{T}$$

$$\vec{L} \parallel \vec{W}$$

So if to think in terms of vectorical math we can write our net force of flying airplane like this:

$$\vec{T} + \vec{D} + \vec{L} + \vec{W} = \vec{F}_{\text{net}}$$

$$\|\vec{F}_{\text{net}}\| = F_{\text{net}}$$

But we are not masachists to write it in vectors if we can write it just using scalars (which is less intuitive, but easier to calculate later). By knowing vector orientations we can form triangles, and use pythagorean theorem:

$$F_{\text{net}} = \sqrt{(T - D)^2 + (L - W)^2}$$

Now we have already a base patter, as for now we need to substitute everything we know:

Therefore, the fully substituted equation is:

$$F_{\text{net}} = \sqrt{\left[T - \frac{1}{2}\rho v^2 S \left(C_{D_p} + \frac{S}{\pi e b^2} \left(\frac{a_0 \alpha}{1 + \frac{a_0 S}{\pi e b^2}} \right)^2 \right) \right]^2 + \left[\frac{1}{2}\rho v^2 S \left(\frac{a_0 \alpha}{1 + \frac{a_0 S}{\pi e b^2}} \right) - mg \right]^2}$$

Also we can change T to equation of force:

$$T = m \frac{\Delta v}{\Delta t}$$

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - \frac{1}{2}\rho v^2 S \left(C_{D_p} + \frac{S}{\pi e b^2} \left(\frac{2\pi \alpha}{1 + \frac{2S}{eb^2}} \right)^2 \right) \right]^2 + \left[\frac{1}{2}\rho v^2 S \left(\frac{2\pi \alpha}{1 + \frac{2S}{eb^2}} \right) - mg \right]^2}$$

Then we substitute:

$$C_{D_p} = 0$$

$$e = 0.7$$

$$a_0 = 2\pi \text{ rad}^{-1}$$

$$C_{L0} = 0$$

and standard near-sea-level values:

$$\rho = 1.225 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

Then:

$$\frac{1}{2}\rho = \frac{1}{2}(1.225) = 0.6125$$

and

$$\frac{a_0 S}{\pi e b^2} = \frac{2\pi S}{\pi(0.7)b^2} = \frac{2S}{0.7b^2} = \frac{20S}{7b^2}$$

equation becomes:

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - 0.6125 v^2 S \left(\frac{S}{0.7\pi b^2} \left(\frac{2\pi\alpha}{1 + \frac{20S}{7b^2}} \right)^2 \right) \right]^2 + \left[0.6125 v^2 S \left(\frac{2\pi\alpha}{1 + \frac{20S}{7b^2}} \right) - 9.81mg \right]^2}$$

Then we need to choose wing span and area. For wing span I will choose units that are wide enough and easy enough to make on 3d printer:

$$b = 0.30 \text{ m}$$

$$S = 0.0225 \text{ m}^2$$

So:

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - 0.6125v^2(0.0225) \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi\alpha}{1.714286} \right)^2 \right) \right]^2 + \left[0.6125v^2(0.0225) \left(\frac{2\pi\alpha}{1.714286} \right) - 9.8 \right]^2}$$

Now we have only 3 variables left, 2 of them we will find in the next section.

2.3 Designing Wide Airplane

I designed an airplane (which you can find in this repo) and found that its weight is approximately 0.0205 kg. Which we can insert into the equation:

$$\sqrt{\left[0.0205 \frac{\Delta v}{\Delta t} - 0.6125v^2(0.0225) \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi\alpha}{1.714286} \right)^2 \right) \right]^2 + \left[0.6125v^2(0.0225) \left(\frac{2\pi\alpha}{1.714286} \right) - 0.201 \right]^2}$$

And we can simplify it further.

Since:

$$0.6125(0.0225) = 0.01378125$$

we get:

$$\sqrt{\left[0.0205 \frac{\Delta v}{\Delta t} - 0.01378125v^2 \left(\frac{0.0225}{0.063\pi} \left(\frac{2\pi\alpha}{1.714286} \right)^2 \right) \right]^2 + \left[0.01378125v^2 \left(\frac{2\pi\alpha}{1.714286} \right) - 0.201105 \right]^2} = F_{\text{net}}$$

Now we need to choose v , α and $\frac{\Delta v}{\Delta t}$.

$$v = 10 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = 50 \text{ m/s}^2$$

Now we need to choose an angle of attack. Since we are making a plastic version of a paper airplane with flat wings, we can use a typical angle of attack of 5° for the model.

Therefore:

$$\alpha = 5^\circ = \frac{5\pi}{180} \approx 0.0873 \text{ rad}$$

First, we calculate the lift:

$$L = 0.01378125(10)^2 \left(\frac{2\pi(0.0873)}{1.714286} \right)$$

$$L \approx 0.44096 \text{ N}$$

Next, we calculate the induced drag:

$$D = 0.01378125(10)^2 \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi(0.0873)}{1.714286} \right)^2 \right)$$

$$D \approx 0.01604 \text{ N}$$

The thrust is:

$$T = (0.0205)(50) = 1.025 \text{ N}$$

and the weight is:

$$W = (0.0205)(9.81) = 0.201105 \text{ N}$$

Therefore, the horizontal and vertical components of the net force are:

$$F_x = T - D = 1.025 - 0.01604 = 1.00896 \text{ N}$$

$$F_y = L - W = 0.44096 - 0.201105 = 0.23986 \text{ N}$$

Finally:

$$F_{\text{net}} = \sqrt{(1.00896)^2 + (0.23986)^2}$$

$$\boxed{F_{\text{net}} \approx 1.037 \text{ N}}$$

In the next section, we will calculate F_{net} for a narrower airplane and compare them.

2.4 Calculation for a Narrower Airplane

The only thing that we will change here is b , which in turn will affect AR . S should stay the same. For b I chose 20 cm. Mass is the same, as i was able to control it with an infill density on a 3d printer. Now what is left to calculate is F_{net} for a narrower airplane:

$$b = 0.20 \text{ m}$$

Therefore:

$$AR = \frac{b^2}{S} = \frac{0.20^2}{0.0225} \approx 1.7778$$

The denominator in the lift coefficient becomes:

$$1 + \frac{2S}{eb^2} = 1 + \frac{2(0.0225)}{0.7(0.20)^2} \approx 2.607143$$

Therefore, the net force equation becomes:

$$F_{\text{net}} = \sqrt{\left[0.0205 \frac{\Delta v}{\Delta t} - 0.6125v^2(0.0225) \left(\frac{0.0225}{0.7\pi(0.04)} \left(\frac{2\pi\alpha}{2.607143}\right)^2\right)\right]^2 + \left[0.6125v^2(0.0225) \left(\frac{2\pi\alpha}{2.607143}\right)\right]^2}$$

Using:

$$v = 10 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = 50 \text{ m/s}^2$$

and:

$$\alpha = 5^\circ = \frac{5\pi}{180} \approx 0.0873 \text{ rad}$$

we get:

$$L = 0.01378125(10)^2 \left(\frac{2\pi(0.0873)}{2.607143}\right)$$

$$L \approx 0.2900 \text{ N}$$

Next, we calculate the induced drag:

$$D = 0.01378125(10)^2 \left(\frac{0.0225}{0.7\pi(0.04)} \left(\frac{2\pi(0.0873)}{2.607143} \right)^2 \right)$$

$$D \approx 0.01858 \text{ N}$$

The thrust remains the same:

$$T = (0.0205)(50) = 1.025 \text{ N}$$

and the weight remains:

$$W = (0.0205)(9.81) = 0.201105 \text{ N}$$

Therefore:

$$F_x = T - D = 1.025 - 0.01858 = 1.00642 \text{ N}$$

$$F_y = L - W = 0.2900 - 0.201105 = 0.08890 \text{ N}$$

Finally:

$$F_{\text{net}} = \sqrt{(1.00642)^2 + (0.08890)^2}$$

$$F_{\text{net}} \approx 1.010 \text{ N}$$